

MAT 1001 Final Exam, 13:30-16:30 am, December 16, 2024

Instruction:

- This is a closed-book and closed-notes exam; no calculators, no dictionaries and no cell phones;
- Check to see if your paper has 6 pages; if not, report it to the proctor in the exam room;
- The full score of the exam is 232 points, to earn which you need to answer ALL questions correctly;
- Write down ALL your work and your answers (including the answers for short questions) in the Exam Book;
- Show your work unless otherwise instructed—a correct answer without showing your work when required shall be given no credits.

1. (54 points) **Short Questions** (for these questions, NO need to show your work, just write down your answers in the Exam Book; NO partial credits for each question)

(i). True or False: If f is differentiable at c , then f is continuous at c .

True

False

(ii). True or False: The derivative of an even and differentiable function is odd, and the derivative of an odd and differentiable function is even.

True

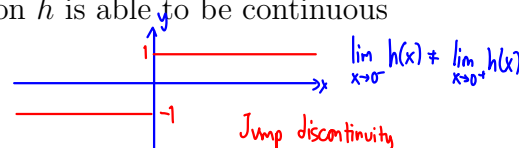
False

$$\begin{aligned} [f(-x) = -f(x)]' & \text{ } f' \text{ odd} \\ -f'(-x) & = -f'(x) \\ f'(-x) & = f'(x) \quad f' \text{ even} \end{aligned}$$

(iii). True or False: Given a function $h(x) = \frac{|x|}{x}$, $x \in (-\infty, 0) \cup (0, \infty)$. By appropriately defining the value of $h(0)$, the extended function h is able to be continuous on $(-\infty, \infty)$.

True

False



$$4y^3 \frac{dy}{dx} - 8y \frac{dy}{dx} = 4x^3 \frac{dx}{dx}$$

(iv). Find the slopes of the devil's curve $y^4 - 4y^2 = x^4 - 9x^2$ at the point $(-3, 2)$.

(v). True or False: Let $f(x) = x^3 + bx^2 + cx + d$ be a cubic function. If $b^2 < 3c$, then f has no local extreme values.

True

False

No c.p. $\rightarrow$ No local extreme

$$\begin{aligned} \frac{dy}{dx} &= \frac{4x^3 - 18x}{4y^3 - 8y} = \frac{-108 + 54}{32 - 16} \\ &= -\frac{27}{8} \end{aligned}$$

$$f'(x) = 3x^2 + 2bx + c$$

$$4b^2 - 12c < 0 \rightarrow \text{Nowhere s.t. } f'(x) = 0$$

4h

$$\left. \begin{array}{l} M = \max \\ m = \min \end{array} \right\} \text{ of } f \text{ on } [a, b]$$

2

- (vi). True or False: If the function f is continuous in $[a, b]$, then there is a point $a < c < b$ such that $f(c) = f(a) + (f(b) - f(a))/3$.

True

False

$$m \leq \frac{2f(a) + f(b)}{3} \leq M \quad \text{IVT: } \exists c \in (x_m, x_M) \text{ s.t. } f(c) = \frac{2f(a) + f(b)}{3}$$

- (vii). Find the linearization $L(x)$ of $x + \frac{1}{x}$ at $x = 1$. $L(x) = f(x) + f'(x_0)(x - x_0)$

$$\begin{aligned} f'(x) &= \frac{1}{x^2} \\ f'(1) &= \frac{1}{1^2} = 1 \end{aligned}$$

- (viii). Let $f(x) = x^3 - 3x^2 - 1$, $x \geq 2$. Find $(f^{-1})'(-1)$. $L(1) = 1 + \frac{1}{1} + (1 - \frac{1}{1})(x - 1) = 2$

- (ix). For the function $f(x) = -2x^3 - x^2 + 6x + 10$, let M be a midpoint Riemann sum of f over the interval $[0, 1]$. Which of the following statements is true? $f'(x) = -6x^2 - 2x + 6$

- (a) $f(x)$ is concave down; $M > \int_0^1 f(x) dx$ always holds no matter what partition of $[0, 1]$ we use. $f''(x) = -12x - 2 < 0$

- (b) $f(x)$ is concave down; $M < \int_0^1 f(x) dx$ always holds no matter what partition of $[0, 1]$ we use.

- (c) None of the above.

- (x). Which of the following functions grows at the fastest rate as $x \rightarrow \infty$?

- (a) $(1314x)^{\pi+2024}$ $\times$ e^{x^2}

- (b) $e^x + 48$ $\times$ $e^{x \ln x}$

- (c) $e^{(x^2)}$ $= (e^x)^x$

- (d) $x^x/3$ $= \frac{1}{3}(x)^x$

- (xi). For the following statements, consider $x \rightarrow \infty$. Choose all that are true.

- (a) $\arctan x = O(1)$ $\lim_{x \rightarrow \infty} \frac{\arctan x}{1} = \frac{\pi}{2}$ $\checkmark$

- (b) $\ln(\ln x) = O(\ln x)$ $\lim_{x \rightarrow \infty} \frac{\ln(\ln x)}{\ln x} = \lim_{u \rightarrow \infty} \frac{\ln u}{u} = 0$

- (c) $x^{-2} + x^{-4} = O(x^{-4})$ $\lim_{x \rightarrow \infty} \frac{x^{-2} + x^{-4}}{x^{-4}} = \lim_{x \rightarrow \infty} x^2 + 1 = \infty$ $\times$

- (d) $x = o(x + \ln x)$ $\lim_{x \rightarrow \infty} \frac{x}{x + \ln x} = \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{\ln x}{x}} = \frac{1}{1+0} = 1$ $\checkmark$

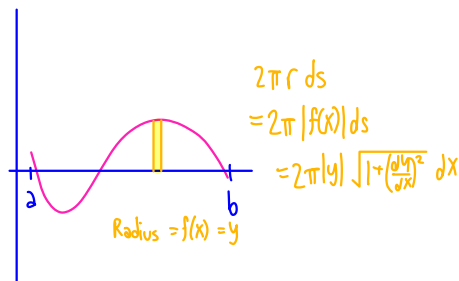
- (xii). Suppose a curve $y = f(x)$, $a \leq x \leq b$, is revolved about the x -axis, the formula that should be used for computing the surface area of the revolution is $\therefore \frac{x}{x + \ln x} \neq o(1)$ $\times$

$$(a) S = 2\pi \int_a^b |y| \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$(b) S = 2\pi \int_a^b |y| \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$(c) S = 2\pi \int_a^b |y| \sqrt{1 + \frac{dy}{dx}} dx$$

$$(d) S = 2\pi \int_a^b |y| \sqrt{1 + \frac{dx}{dy}} dy$$



- (xiii). The volume of the solid generated by rotating about the x -axis the region below the curve $y = \sqrt{x}$ and above the interval $[0, 1]$ is

$$= \int_0^1 \pi (\sqrt{x})^2 dx = \pi \left[\frac{x^2}{2} \right]_0^1 = \frac{\pi}{2}$$

- (a) $\frac{\pi}{2}$
 (b) $\frac{\pi}{3}$
 (c) 2π
 (d) π^2

$$-1 = \left[(x-1)e^{x^2} \right]_0^1 - \int_0^1 v dv$$

(xiv). True or False: If $uv = (x-1)e^{x^2}$ and $\int_0^1 u dv = -1$, then $\int_0^1 v du = 0$.

True

False

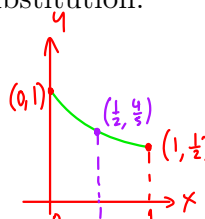
(xv). To find the integral $\int \sec^4 \theta d\theta$, we use the substitution:

(a) $u = \sin \theta$

(b) $u = \cos \theta$

(c) $u = \tan \theta$

$$\begin{aligned} &= \int \sec^2 \theta (\sec^2 \theta d\theta) \\ &= \int (\tan^2 \theta + 1) (\sec^2 \theta d\theta) \\ &= \int (u^2 + 1) du \end{aligned}$$



$$S_n = \frac{1}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + \dots + 4y_{n-1} + y_n) \Delta x$$

$$\begin{aligned} S_n &= \frac{1}{3}(y_0 + 4y_1 + y_2) \cdot \Delta x \\ &= \frac{1}{3}\left(1 + 4\left(\frac{1}{2}\right) + \frac{1}{2}\right) \cdot \frac{1}{2} \\ &= \frac{47}{60} \end{aligned}$$

(xvi). Estimate $I = \int_0^1 \frac{1}{1+x^2} dx$ by using Simpson's Rule with $n = 2$ steps.

(xvii). Use Euler's method to approximate $y(3)$, where $y(x)$ is the solution of the following initial value problem

$$y' = 1 - \frac{y}{x}, \quad y(2) = -1,$$

$$\begin{aligned} y(2.5) &= y(2) + y'(2) \Delta x \\ &= -1 + \frac{3}{2} \cdot 0.5 \end{aligned}$$

with the increment size $dx = \Delta x = 0.5$.

$$\begin{aligned} y(3) &= y(2.5) + y'(2.5) \Delta x \\ &= -0.25 + 1.1 \cdot 0.5 = 0.3 \end{aligned}$$

(xviii). Which of the following statements is NOT correct?

(a) $dH/dt = H_S - H$ can be used to model Newton's Law of Cooling, where $H(t)$ is the temperature of an object at time t and H_S is the constant temperature of the surrounding medium. $\checkmark$ ($k=1$)

(b) The criticism on the Malthusian Model (Exponential Population Growth Model) $dP/dt = kP$ (k is a positive constant) is that the exponential growth of the population predicted by this model cannot be sustained in the real world, due to limited number of resources. $\checkmark$

(c) $dP/dt = P^2 - P$ is a special case of the logistic model. $\times$

$$\begin{aligned} \frac{1}{P^2 - P} dP &= \frac{1}{dt} dt & \ln\left|\frac{P-1}{P}\right| &= t + C \\ \int \frac{1}{P^2 - P} dP &= t & \frac{1-P}{P} &= ke^t \\ \frac{1}{P} - \frac{1}{P-1} &= t & \frac{1}{P} - 1 &= ke^t \\ P &= \frac{1}{1+ke^t} \end{aligned}$$

2. (24 points) Find each of the following limits or explain why the limit does not exist.

(a) $\lim_{x \rightarrow -\infty} \sqrt{x^2 + 2024} + x = \lim_{x \rightarrow -\infty} \frac{2024}{\sqrt{x^2 + 2024} - x} = \lim_{x \rightarrow -\infty} \frac{2024}{-x(\sqrt{1 + \frac{2024}{x^2}} + 1)} = 0$

(b) $\lim_{x \rightarrow 0} \frac{\sin(3x) - 3x + x^2}{\sin(x) \sin(2x)} \rightarrow \frac{3\cos 3x - 3 + 2x}{2\sin x \cos 2x + \sin 2x \cos x} \rightarrow \frac{-9\sin 3x + 2}{2(-2\sin x \sin 2x + \cos x \cos 2x) + (-\sin 2x \sin x + 2\cos 2x \cos x)} = -\frac{9\sin 3x + 2}{4} = \frac{2}{4} = \frac{1}{2}$

(c) $\lim_{x \rightarrow 0^+} [\cos(\sqrt{x})]^{1/\sin x} = \lim_{x \rightarrow 0^+} e^{\frac{\ln(\cos \sqrt{x})}{\sin x}} = e^{-\frac{1}{2}}$

(d) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n e^{\frac{k}{n}} = \frac{\frac{1}{\cos \sqrt{x}} \cdot (-\sin \sqrt{x}) \cdot \frac{1}{2\sqrt{x}}}{\cos x} = \frac{\frac{1}{1} \cdot \left(-\frac{\sin \sqrt{x}}{2\sqrt{x}}\right)}{1} = \frac{1 \cdot \left(-\frac{1}{2}\right)}{1} = -\frac{1}{2}$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \frac{1}{n} \left(e^{\frac{1}{n}} + e^{\frac{2}{n}} + \dots + e^{\frac{n}{n}} \right) \\ &= \int_0^1 e^x dx = [e^x]_0^1 = e - 1 \end{aligned}$$

3. (18 points) Find each of the following derivatives or explain why the derivative does not exist.

(a) $\frac{d}{dx} x^{\sin x} = \frac{d}{dx} e^{\sin x \cdot \ln x} = x^{\sin x} \cdot \left(\frac{\sin x}{x} + \ln x \cos x \right)$

Recall: $\frac{d}{dx} \int_a^x f(t) dt = f(x)$

(b) $f'(x)$, where $f(x) = \begin{cases} x^2-1 & ; x \geq 1 \text{ or } x \leq -1 \\ 1-x^2 & ; -1 < x < 1 \end{cases}$

$f(x) = \int_{x^4}^{x^2} \log_{10}(\sqrt{u}) du = \int_1^{x^2} \log_{10} \sqrt{u} du - \int_1^{x^4} \log_{10} \sqrt{u} du$

$f'(x) = \left[\log_{10} \sqrt{u} \right]_{u=x^2} - \left[\log_{10} \sqrt{u} \right]_{u=x^4}$

$= \log_{10} |x| - \log_{10} x^2$

DNE (c) $f'(1)$, where $f(x) = |x^2 - 1|$. $f'(x) = -2$, $f'(x) = 2$

4. (16 points) Let $f(x) = (x^2 + 2x + 2)e^{-x}$.

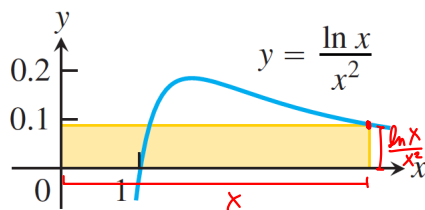
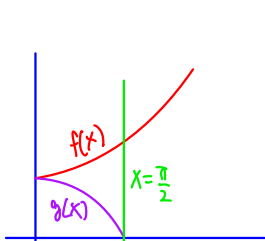
(a) Find the open intervals on which f is increasing and decreasing.

(b) Find the coordinates of its local maximums and local minimums (if any).

(c) Find the open intervals on which f is concave up and concave down.

(d) Find the coordinates of its inflection points (if any).

5. (8 points) The rectangle shown below has one side on the positive y -axis, one side on the positive x -axis, and its upper-right vertex on the curve $y = (\ln x)/x^2$, $x \geq 1$. What value of x gives the rectangle its largest area, and what is that area?



Max of $f(x)$

$f(x) = \frac{\ln x}{x^2}$ $\ln x \cdot \frac{1}{x^2} \Rightarrow \ln x \cdot -\frac{1}{x^2} + \frac{1}{x^2}$

$\rightarrow f'(x) = \frac{1 - \ln x}{x^2} \Rightarrow f'(x) = 0$ when $x = e$

$\rightarrow f(x) = \frac{1}{e}$ $\leftarrow$ maximum

6. (8 points) Find the area of the region in the first quadrant bounded by the graphs of the functions $f(x) = e^x$, $g(x) = \cos x$ and the line $x = \pi/2$.

7. (8 points) Let R be the (finite) region bounded by $y = (x-1)^2$ and $y = x+1$. Find the volume of the solid obtained by rotating R about the x -axis.

8. (36 points) Compute the following integrals

(i)

$u = \sin \sqrt{x} \rightarrow du = \cos \sqrt{x} \cdot \frac{1}{2\sqrt{x}} dx$

$\int \frac{\cos \sqrt{x}}{\sqrt{x} \sin \sqrt{x}} dx$

$= \int \frac{2 du}{u} = 2 \ln |u| + C = 2 \ln |\sin \sqrt{x}| + C$

(ii)

$\int \frac{dx}{(x+3)(x^2+2x+1)}$

$= \int \frac{A}{x+3} + \frac{B}{x+1} + \frac{C}{(x+1)^2} dx = \frac{1}{4} \ln |x+3| - \frac{1}{4} \ln |x+1| - \frac{1}{2(x+1)} + C$

$1 = A(x^2+2x+1) + B(x^2+4x+3) + C(x+3)$

$A+B=0$

$2A+4B+C=0$

$A+3B+3C=1$

$2B+C=0$

$2B+3C=1$

$B=-\frac{1}{4} \rightarrow A=\frac{1}{4}$

$C=\frac{1}{2}$

$1 = A(x+1)^2 + B(x+1)(x+3) + C(x+3)$

$x=-3 \rightarrow A=\frac{1}{4}, x=-1 \rightarrow C=\frac{1}{2}$

$1 = \frac{1}{4} \cdot 1 + B \cdot 3 + \frac{1}{2} \cdot 3 \rightarrow 3B = -\frac{3}{4} \therefore B = -\frac{1}{4}$

(iii)

$$\int \frac{dx}{\sqrt{4x-x^2}} = \int \frac{dX}{\sqrt{4-(X+2)^2}} = \int \frac{dU}{\sqrt{4-U^2}} = \int \frac{2dV}{\sqrt{4-4V^2}} = \int \frac{dV}{\sqrt{1-V^2}} = \arcsin(V) + C = \arcsin\left(\frac{X}{2}+1\right) + C$$

$U=X+2 \rightarrow dU=dX$
 $V=\frac{U}{2} \rightarrow dV=\frac{1}{2}dU$

$$\begin{aligned} \text{(iv)} \quad u &= \ln(x+1) \quad v = \frac{x^2}{2} \\ &= \frac{x^2}{2} \ln(x+1) - \int \frac{x^2}{2(x+1)} dx \\ &= \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \int \left(x - \frac{1}{x+1} \right) dx \end{aligned}$$

$$\int x \ln(x+1) dx$$

$$\int \frac{dX}{\sqrt{4-X^2}} = \arcsin\left(\frac{X}{2}\right) + C$$

$$\text{(v)} \quad = \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \left(\frac{x^2}{2} - x - \ln|x+1| \right) + C$$

$$\begin{aligned} u &= e^x \rightarrow du = e^x dx & u &= 2\sec\theta \rightarrow du = 2\sec\theta \tan\theta d\theta; \theta \in [0, \frac{\pi}{2}) \\ \int \frac{e^x dx}{\sqrt{e^{2x}-4}} &= \int \frac{du}{\sqrt{u^2-4}} = \int \frac{2\sec\theta \tan\theta d\theta}{2|\tan\theta|} = \int \sec\theta d\theta = \ln|\sec\theta + \tan\theta| + C \\ &= \ln\left(\frac{e^x}{2} + \sqrt{\frac{e^{2x}}{4}-1}\right) + C \end{aligned}$$

$\sec\theta > 0$
 $\sec X, \tan X > 0$ if $X \in [0, \frac{\pi}{2})$

(vi)

$$\int \sin(2x) \sin(6x) dx = \int \frac{1}{2} (\cos 4x - \cos 8x) dx = \frac{1}{2} \left(\frac{1}{4} \sin 4x - \frac{1}{8} \sin 8x \right) + C$$

9. (10 pts) The graph of $y = x^3$ on $0 \leq x \leq 1$ is revolved about the y -axis to form a tank that is then filled with salt water from the Dead Sea (density is approximately 11400 N/m^3), where both x and y are measured in meters. Set up an integral for the work needed to pump all the water to the top of the tank. **Do not spend your time computing the integral!**
- $y = x^3$
 $x = y^{1/3}$

10. (12 points) Determine the convergence of each of the following improper integrals (you are not required to find its value if the integral is convergent).

(i)

$$\int_1^{\infty} \frac{2 + \sin x}{x} dx = \int_1^{\infty} \frac{2}{x} dx + \int_1^{\infty} \frac{\sin x}{x} dx \rightarrow \text{diverge}$$

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \rightarrow \sin \theta \approx \theta \text{ when } \theta \rightarrow 0$$

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sin \sqrt{x}} = 1$$

$$\sin \theta = \theta + o(\theta); \theta \rightarrow 0 \rightarrow \int_0^1 \frac{1}{\sin \sqrt{x}} dx \text{ and } \int_0^1 \frac{1}{\sqrt{x}} dx \text{ conv or div at the}$$

$$(8.8/41) \sin \theta < \theta$$

$$\int_0^1 \frac{1}{\sin(\sqrt{x})} dx$$

Converges

$$\because \sin \sqrt{x} > \frac{\sqrt{x}}{2}$$

$$\frac{1}{\sin \sqrt{x}} < \frac{2}{\sqrt{x}}$$

$$\int_0^1 \frac{1}{\sin \sqrt{x}} dx < \int_0^1 \frac{2}{\sqrt{x}} dx = [4\sqrt{x}]_0^1 = 4$$

11. (12 pts) Solve the following differential equations

$$\text{(a)} \quad \frac{dy}{dx} = xe^{x^2-y};$$

$$\frac{dy}{dx} = \frac{xe^x}{e^y} \rightarrow \int e^y dy = \int xe^x dx$$

$$e^y = \frac{e^x}{2} + C$$

$$\sin x = \frac{x}{2}$$

$$\cos x = \frac{1}{2}$$

$$\frac{2y}{3} > 1$$

$$\rightarrow \cos x \geq \cos 1 > \cos \frac{2\pi}{3} = -\frac{1}{2}$$

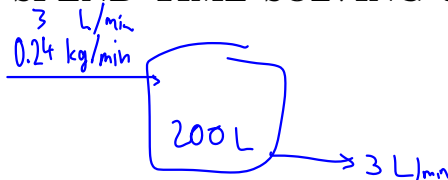
$$\text{(b)} \quad \frac{dy}{dx} - 2y = x^2, \quad x > 0, \quad y(1) = 2.$$

$$v(x) = e^{\int -2 dx} = e^{-2\ln x} = \frac{1}{x^2}$$

$$y = \frac{1}{x^2} \int x^2 \cdot \frac{1}{x^2} dx = x^2 (x+C) = x^3 + Cx^2 = x^3 + x^2$$

$$y(1)=2 \rightarrow 2=1+C \rightarrow C=1$$

12. (10 points) A tank initially contains 200 L of fresh water. A solution containing 0.08 kg/L of soluble lawn fertilizer runs into the tank at the rate of 3 L/min, and the well-mixed mixture is pumped out of the tank at the same rate. **Derive** a differential equation for the amount $F(t)$ of fertilizer inside the tank at time t , and specify the initial condition $F(0)$. **DO NOT SPEND TIME SOLVING the DIFFERENTIAL EQUATION!**



$$\frac{dF}{dt} = 0.24 - \frac{F}{200} \cdot 3$$

$$\rightarrow F'(t) + \frac{3}{200}F = 0.24, \quad F(0) = 0 \text{ (Freshwater)}$$

13. (16 points) Consider the autonomous differential equation

$$\frac{dP}{dt} = 3P(9 - P)(P - 0.07),$$

where $P(t)$, measured in thousands, is the number of graduate students at CUHKSZ at time t .

- (i) Find all equilibrium solutions of the differential equation.

- (ii) Perform phase-line analysis to the equation.

- (iii) We know that currently CUHKSZ has 4.1 thousands graduate students. Use the result from (ii) to predict the long-time population of graduate students on the campus, i.e., predict $\lim_{t \rightarrow \infty} P(t)$.

- (iv) What would be the long-time population of graduate students if the university started off with only 60 graduate students?

$$\lim_{t \rightarrow \infty} P(t) = 0$$

(Gone)

$$i) \int_0^{\pi/2} \frac{\cos t}{t + \sin t} dt$$

$$\therefore \lim_{t \rightarrow 0} \frac{\frac{\cos t}{t + \sin t}}{\frac{1}{2t}} = \lim_{t \rightarrow 0} \frac{2t \cos t}{t + \sin t} = \lim_{t \rightarrow 0} \frac{2 \cos t}{1 + \frac{\sin t}{t}} = \frac{2 \cdot 1}{1 + 1} = 1$$

$$\therefore \int_0^{\pi/2} \frac{1}{2t} dt \text{ diverge} \rightarrow \int_0^{\pi/2} \frac{\cos t}{t + \sin t} dt \text{ diverge}$$

$$ii) \int_{-\infty}^{\infty} \frac{2X}{(X^2+1)^{3/2}} dX = \int_{-\infty}^0 + \int_0^{\infty}$$

$$\int_0^{\infty} \frac{2X}{(X^2+1)^{3/2}} dX = \lim_{a \rightarrow \infty} \int_0^a \frac{2X}{(X^2+1)^{3/2}} dX = \lim_{a \rightarrow \infty} \left[-2(X^2+1)^{-1/2} \right]_0^a = -2(0-1) = 2$$

$$\int_{-\infty}^0 \frac{2X}{(X^2+1)^{3/2}} dX = - \int_0^{\infty} \frac{2X}{(X^2+1)^{3/2}} dX \quad (\text{Odd func}) = -2$$

$$\therefore \int_{-\infty}^{\infty} \frac{2X}{(X^2+1)^{3/2}} dX = -2 + 2 = 0$$

$$3. i) \frac{dy}{dx} = e^{x+y} = e^x \cdot e^y$$

$$e^{-y} dy = e^x dx$$

$$\int e^{-y} dy = \int e^x dx$$

$$-e^{-y} = e^x + C_1$$

$$e^{-y} + e^x = C$$

$$-y = \ln(C - e^x)$$

$$y = -\ln(C - e^x)$$

$$ii) x \frac{dy}{dx} - 2y = x^3 \sec x \tan x \quad (y(\frac{\pi}{3}) = 2)$$

$$\frac{dy}{dx} - \frac{2y}{x} = x^2 \sec x \tan x$$

$$I = e^{\int -\frac{2}{x} dx} = \frac{1}{x^2}$$

$$(Iy)' = \cancel{x^2} \sec x \tan x \cdot I \cdot \cancel{\frac{1}{x^2}}$$

$$Iy = \int \sec x \tan x dx = \sec x + C$$

$$y = x^2 \sec x + Cx^2$$

$$2 = \frac{\pi^2}{9} (2) + \frac{\pi^2}{9} \cdot C$$

$$\frac{18}{\pi} = 2 + C \rightarrow C = \frac{18}{\pi} - 2$$